ELEC4631 Tutorial 9

1. Let $X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \in \mathbb{R}^{2\times 2}$ and $Y = \begin{bmatrix} y_{11} & y_{21} \\ y_{21} & y_{22} \end{bmatrix} \in \mathrm{sym}(\mathbb{R}^{2\times 2})$. Consider the affine map $L : \mathbb{R}^{2\times 2} \times \mathrm{sym}(\mathbb{R}^{2\times 2}) \to \mathrm{sym}(\mathbb{R}^{2\times 2})$ given by

$$\begin{aligned} L(X,Y) &= \begin{bmatrix} -1 & 3 \\ 3 & -4 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} X + X^T \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} \\ &\quad + \begin{bmatrix} -1 & 2 \\ 3 & -4 \end{bmatrix} Y \begin{bmatrix} -3 & 1 \\ 3 & 9 \end{bmatrix} + \begin{bmatrix} -3 & 3 \\ 1 & 9 \end{bmatrix} Y \begin{bmatrix} -1 & 3 \\ 2 & -4 \end{bmatrix}. \end{aligned}$$

   Put the strict LMI $L(X,Y) > 0$ into a single standard form LMI in the decision variable $x = (x_{11}, x_{21}, x_{12}, x_{22}, y_{11}, y_{21}, y_{22})^T$.

2. Using the Schur complement, solve the following LMIs and sketch the set of of all vectors x that satisfy it (if the LMI is feasible)

   (a) The LMI from Problem 4 of Tutorial 9.

   (b)
$$\begin{bmatrix} 3x_1 + 2x_2 & x_2 \\ x_2 & x_1 + x_2 \end{bmatrix} > 0.$$

3. Let $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$ be given constant matrices. Let $X = X^T \in \mathbb{R}^{n\times n}$ and $K \in \mathbb{R}^{m\times n}$ be matrix variables. Show that by a suitable change of variables the pair of inequalities

$$\begin{aligned} (A - BK)X(A - BK)^T - X &< 0, \\ X &> 0, \end{aligned}$$

   can be transformed into a strict LMI. Also, show how X and K can be recovered from the solution of the LMI (if it is strictly feasible).

4. If an asymptotically stable linear system has a convergence rate $\mu > 0$ then all its eigenvalues will have a real part less than or equal to $-\mu$. In the lectures it was shown, using Grönwall's inequality, that $\mu > 0$ is a convergence rate for the system $\dot{x}(t) = Ax(t)$ if the LMI

$$\begin{aligned} P &> 0, \\ A^T P + PA &\leq -2\mu P \end{aligned} \tag{1}$$

is feasible.

In this exercise you will show that a slightly restricted version of the converse is also true. Suppose that a linear system has convergence rate larger than μ. Show that the LMI (1) must then be strictly feasible.

5. Suppose that a linear time-invariant state-space system is controllable and observable, but only the output $y(t)$ is accessible rather than the state $x(t)$. Consider the design of a linear output feedback controller using an observer (Lecture 8). In theory it is possible to make the system state go to 0 as fast as desired and also for the observer state to converge to the system state as fast desired. However, this is an idealisation and in physical systems it is not possible to have them respond as fast as one would like. Thus, rather than only specifying a minimum convergence rate $\mu_{\mathrm{s,min}}$ for the system and $\mu_{\mathrm{o,min}}$ for the observer, based on physical characteristics of the plant and the implementation of the observer it is meaningful to also set a maximum convergence rate $\mu_{\mathrm{s,max}} > 0$ and $\mu_{\mathrm{o,max}} > 0$ for the system and observer, respectively, to respect physical limits.

   (a) Formulate the design of the linear output feedback controller (finding the controller gain K and observer gain L) so that $\mu_{\mathrm{s,min}} < \mu_{\mathrm{s}} < \mu_{\mathrm{s,max}}$ and $\mu_{\mathrm{o,min}} < \mu_{\mathrm{o}} < \mu_{\mathrm{o,max}}$ using LMIs, where μ_{s} denotes the system convergence rate and μ_{o} the observer convergence rate. (In general, feasibility of the LMIs will only be sufficient to guarantee the desired convergence rates but need not be necessary. That is, if the LMIs turn out to be infeasible it does not necessarily mean that one cannot design a controller that meets the requirements, it only means that this particular approach does not give a solution).

   (b) Consider a system with

   $$A = \begin{bmatrix} 0 & 0.01 \\ 0 & 0.99 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; \; C = [\; 0.01 \quad 1 \;], \; D = 0,$$

   use Matlab to design a state feedback controller and observer such that $1 < \mu_{\mathrm{s}} < 2$ and $3 < \mu_{\mathrm{o}} < 4$ by solving the LMIs from part (a).